Dhirubhai Ambani University
Technology
Formerly DA-IICT

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Gandhinagar, India

Database ERD & Relational Schema

Course: IT214-Database management system
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Project Team

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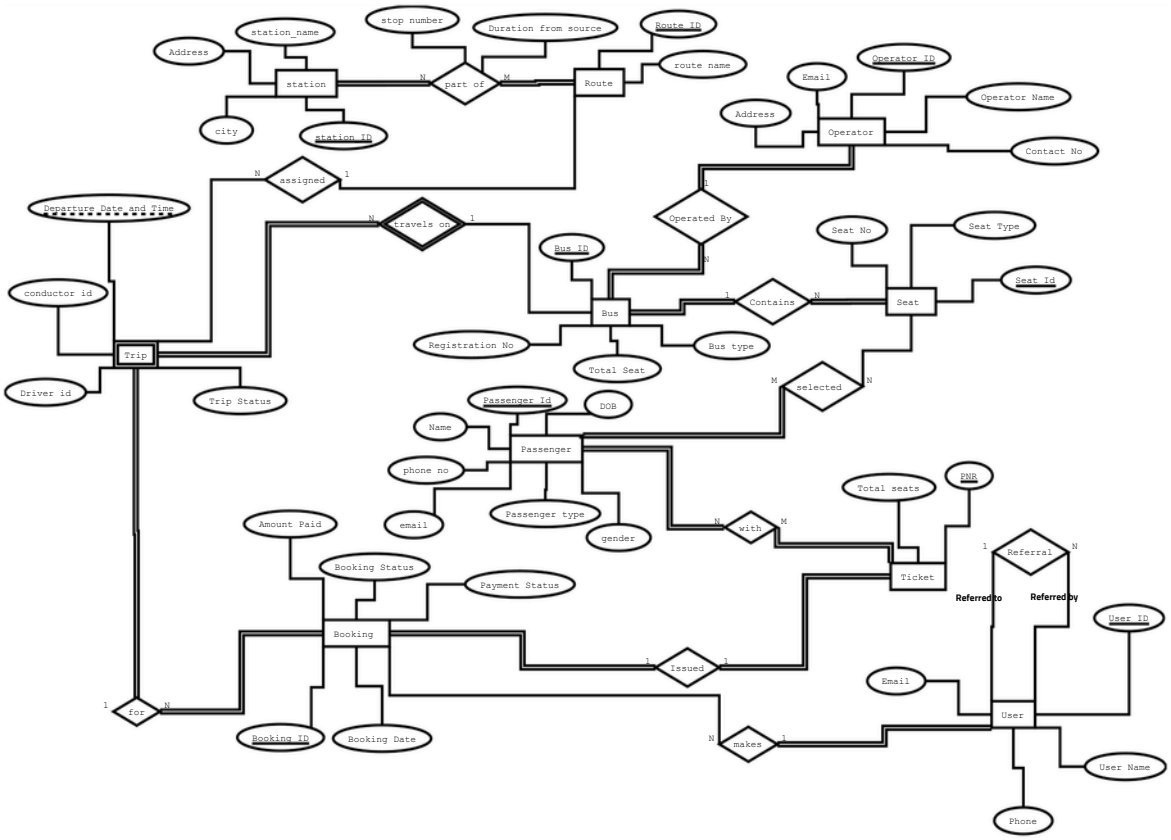
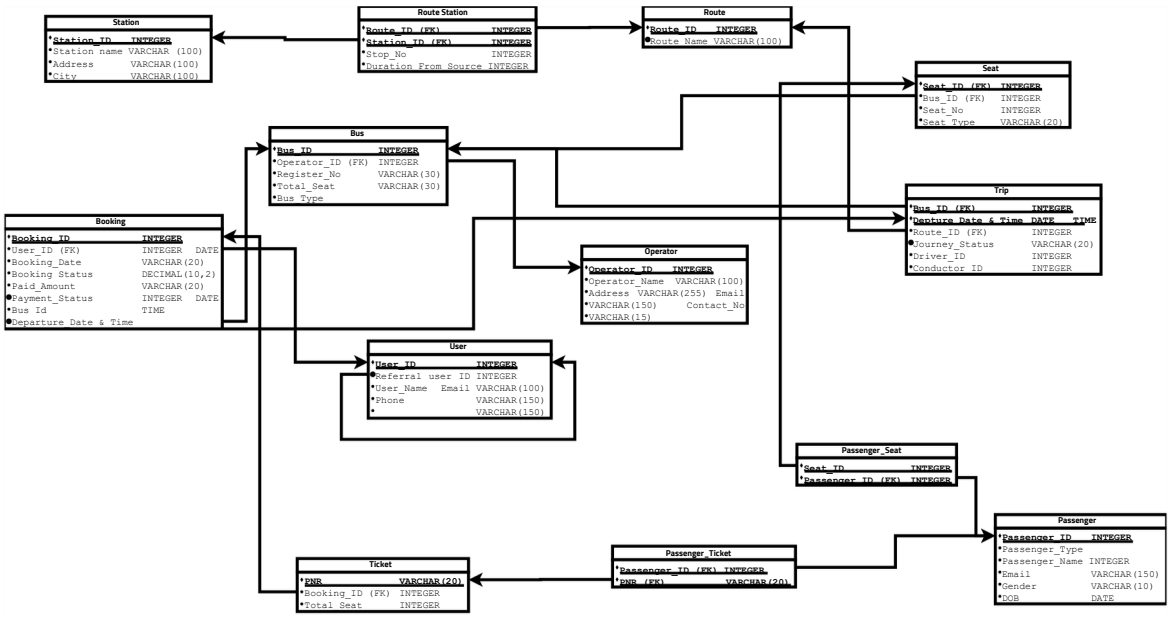
Updated ER Diagram




Station
Station_ID INTEGER
Station name VARCHAR (100)
Address VARCHAR(100)
City VARCHAR(100)
Route Station
Route_ID (FK) INTEGER
Station_ID (FK) INTEGER
Stop_No INTEGER
Duration From Source INTEGER
Route
Route_ID INTEGER
Route Name VARCHAR(100)
Seat
Seat_ID (FK) INTEGER
Bus_ID (FK) INTEGER
Seat_No INTEGER
Seat Type VARCHAR(20)
Bus
Bus_ID INTEGER
Operator_ID (FK) INTEGER
Register_No VARCHAR(30)
Total_Seat VARCHAR(30)
Bus Type
Trip
Bus_ID (FK) INTEGER
Departure Date & Time DATE TIME
Route_ID (FK) INTEGER
Journey_Status VARCHAR(20)
Driver_ID INTEGER
Conductor ID INTEGER
Booking
Booking_ID INTEGER
User_ID (FK) INTEGER DATE
Booking_Date VARCHAR(20)
Booking Status DECIMAL(10,2)
Paid_Amount VARCHAR(20)
Payment_Status INTEGER DATE
Bus Id TIME
Departure Date & Time
Operator
Operator_ID INTEGER
Operator_Name VARCHAR(100)
Address VARCHAR(255) Email
VARCHAR(150) Contact_No
VARCHAR(15)
User
User_ID INTEGER
Referral user ID INTEGER
User_Name Email VARCHAR(100)
Phone VARCHAR(150)
VARCHAR(150)
Passenger_Seat
Seat_ID INTEGER
Passenger_ID (FK) INTEGER
Passenger
Passenger_ID INTEGER
Passenger_Type
Passenger_Name INTEGER
Email VARCHAR(150)
Gender VARCHAR(10)
DOB DATE
Ticket
PNR VARCHAR(20)
Booking_ID (FK) INTEGER
Total Seat INTEGER
Passenger_Ticket
Passenger_ID (FK) INTEGER
PNR (FK) VARCHAR(20)

GoBus — Minimal FD Set

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1) Bus
- BusID → RegNo
- BusID → TotalSeats
- BusID → BusType
- BusID → OperatorID
- RegNo → BusID

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2) Operator
- OperatorID → Operator_Name
- OperatorID → Address
- OperatorID → Email
- OperatorID → ContactNo
- Email → OperatorID
- ContactNo → OperatorID

---

3) Route ● RouteID → RouteName ● RouteName → RouteID

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4) Station
- StationID → StationName
- StationID → Address
- StationID → City
- {StationName, City} → StationID

---

5) RouteStation
- {RouteID, StopNo} → StopNo
- {RouteID, StationID} → DurationFromSource
- {RouteID, StopNo} → StationID

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6) Seat
- SeatID → BusID
- SeatNo → SeatNo
- SeatID → SeatType
- {BusID, SeatNo} → SeatID

7) User ● UserID → ReferralID ● UserID → Username ● UserID → Email ● UserID → Phone ● Email → UserID ● Phone → UserID

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8) Passenger

- PassengerID → PassengerType
- PassengerID → PassengerName
- PassengerID → Email
- PassengerID → Gender
- PassengerID → DOB

---

9) Ticket

- PNR → BookingID
- PNR → TotalSeat
- BookingID → PNR

---

10) Booking

- BookingID → UserID
- BookingID → BookingDate
- BookingID → BookingStatus
- BookingID → PaidAmount
- BookingID → PaymentStatus
- BookingID → BusID
- BookingID → DepartureDateAndTime

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11) Trip

- {BusID, DepartureDateAndTime} → RouteID
- {BusID, DepartureDateAndTime} → JourneyStatus
- {BusID, DepartureDateAndTime} → DriverID
- {BusID, DepartureDateAndTime} → ConductorID

BCNF Check — GoBus

Rule: For every functional dependency X → Y, X must be a Superkey of the relation.

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1) Bus

Attributes: BusID,RegNo,TotalSeats, BusType, OperatorID

Finding Candidate Keys:

- BusID → RegNo, TotalSeats, BusType,OperatorID→BusID determines all attributes → **BusID is a Candidate Key**
- RegNo → BusID →BusID→ RegNo,TotalSeats,BusType, OperatorID → RegNo determines allattributes→ **RegNo is a Candidate Key**

Candidate Keys: {BusID}, {RegNo}

Checking each FD for BCNF:

- BusID → RegNo → BusID is a Candidate Key
- BusID → TotalSeats → BusID is a Candidate Key
- BusID → BusType → BusID is a Candidate Key
- BusID → OperatorID → BusID is a Candidate Key
- RegNo → BusID → RegNo is a Candidate Key

Bus is in BCNF

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2) Operator

Attributes: OperatorID,Operator_Name, Address, Email, ContactNo

Finding Candidate Keys:

- OperatorID → Operator_Name,Address, Email,ContactNo → OperatorID determines all attributes→ **OperatorID is a Candidate Key**
- Email→OperatorID→OperatorID→all attributes→Email determines all attributes → **Email is a Candidate Key**
- ContactNo→OperatorID→OperatorID →allattributes→ ContactNo determines all attributes → **ContactNo is a Candidate Key**

Candidate Keys: {OperatorID}, {Email}, {ContactNo}

Checking each FD for BCNF:

- OperatorID → Operator_Name → OperatorID is a Candidate Key
- OperatorID → Address → OperatorID is a Candidate Key
- OperatorID → Email → OperatorID is a Candidate Key
- OperatorID → ContactNo → OperatorID is a Candidate Key
- Email → OperatorID → Email is a Candidate Key
- ContactNo → OperatorID → ContactNo is a Candidate Key

Operator is in BCNF

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3) Route

Attributes: RouteID,RouteName

Finding Candidate Keys:

- RouteID→ RouteName → RouteIDdetermines all attributes → **RouteID is a Candidate Key**
- RouteName →RouteID → RouteName determines all attributes→ **RouteName is a Candidate Key**

Candidate Keys: {RouteID}, {RouteName}

Checking each FD for BCNF:

- RouteID → RouteName → RouteID is a Candidate Key
- RouteName → RouteID → RouteName is a Candidate Key

Route is in BCNF

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4) Station

Attributes: StationID,StationName, Address, City

Finding Candidate Keys:

- StationID → StationName, Address, City → StationID determines all attributes → **StationID is a Candidate Key**
- {StationName, City} → StationID → StationID → StationName,Address,City → {StationName, City} determines all attributes → **{StationName, City} is a Candidate Key**

Candidate Keys: {StationID}, {StationName, City}

Checking each FD for BCNF:

- StationID → StationName → StationID is a Candidate Key
- StationID → Address → StationID is a Candidate Key
- StationID → City → StationID is a Candidate Key
- {StationName, City} → StationID → {StationName, City} is a Candidate Key

Station is in BCNF

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5) RouteStation

Attributes: RouteID, StopNo, StationID, DurationFromSource

Finding Candidate Keys:

- {RouteID, StopNo} → StationID, DurationFromSource → {RouteID, StopNo} determines all attributes → {RouteID, StopNo} is a Candidate Key
- {RouteID, StationID} → StopNo, DurationFromSource → {RouteID, StationID} determines all attributes → {RouteID, StationID} is a Candidate Key (Reasoning: A specific station on a specific route is assigned exactly one stop number)

Candidate Keys: {RouteID, StopNo}, {RouteID, StationID}

Checking each FD for BCNF:

- {RouteID, StopNo} → StationID → {RouteID, StopNo} is a Candidate Key
- {RouteID, StopNo} → DurationFromSource → {RouteID, StopNo} is a Candidate Key
- {RouteID, StationID} → StopNo → {RouteID, StationID} is a Candidate Key
- {RouteID, StationID} → DurationFromSource → {RouteID, StationID} is a Candidate Key

RouteStation is in BCNF

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6) Seat

Attributes: SeatID, BusID, SeatNo, SeatType

Finding Candidate Keys:

- {BusID, SeatNo} → SeatID, SeatType → {BusID, SeatNo} determines all attributes → {BusID, SeatNo} is a Candidate Key SeatID → BusID, SeatNo, SeatType → SeatID
- determines all attributes → SeatID is a Candidate Key (Reasoning: A primary key ID perfectly identifies every detail of the row, including the seat number)

Candidate Keys: {SeatID}, {BusID, SeatNo}

Checking each FD for BCNF:

- {BusID, SeatNo} → SeatID → {BusID, SeatNo} is a Candidate Key
- {BusID, SeatNo} → SeatType → {BusID, SeatNo} is a Candidate Key
- SeatID → BusID → SeatID is a Candidate Key
- SeatID → SeatNo → SeatID is a Candidate Key
- SeatID → SeatType → SeatID is a Candidate Key

Seat is in BCNF

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7) User

Attributes: UserID,ReferralID, Username, Email, Phone

Finding Candidate Keys:

- UserID → ReferralID, Username, Email, Phone → UserID determines all attributes → **UserID is a Candidate Key**
- Email → UserID→ UserID→all attributes → Email determines all attributes → **Email is a Candidate Key**
- Phone → UserID→ UserID→all attributes → Phone determines all attributes → **Phone is a Candidate Key**

Candidate Keys: {UserID}, {Email}, {Phone}

Checking each FD for BCNF:

● UserID → ReferralID → UserID is a Candidate Key ● UserID → Username → UserID is a Candidate Key ● UserID → Email → UserID is a Candidate Key ● UserID → Phone → UserID is a Candidate Key ● Email → UserID → Email is a Candidate Key ● Phone → UserID → Phone is a Candidate Key

User is in BCNF

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8) Passenger

Attributes: PassengerID,PassengerType, PassengerName, Email, Gender, DOB

Finding Candidate Keys:

- PassengerID → PassengerType, PassengerName, Email, Gender, DOB → PassengerID determines all attributes → **PassengerID is a Candidate Key**

Candidate Keys: {PassengerID}

Checking each FD for BCNF:

- PassengerID → PassengerType → PassengerID is a Candidate Key
- PassengerID → PassengerName → PassengerID is a Candidate Key
- PassengerID → Email → PassengerID is a Candidate Key
- PassengerID → Gender → PassengerID is a Candidate Key
- PassengerID → DOB → PassengerID is a Candidate Key

Passenger is in BCNF

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9) Ticket

Attributes: PNR, BookingID, TotalSeat

Finding Candidate Keys:

- PNR → BookingID, TotalSeat → PNR determines all attributes → **PNR is a Candidate Key**
- BookingID → PNR → PNR → BookingID, TotalSeat → BookingID determines all attributes → **BookingID is a Candidate Key**

Candidate Keys: {PNR}, {BookingID}

Checking each FD for BCNF:

- PNR → BookingID → PNR is a Candidate Key
- PNR → TotalSeat → PNR is a Candidate Key
- BookingID → PNR → BookingID is a Candidate Key

Ticket is in BCNF

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10) Booking

Attributes: BookingID, UserID, BookingDate, BookingStatus, PaidAmount, PaymentStatus, BusID, DepartureDateAndTime
Finding Candidate Keys:

- BookingID → UserID, BookingDate, BookingStatus, PaidAmount, PaymentStatus, BusID, DepartureDateAndTime → BookingID determines all attributes → **BookingID isa CandidateKey**

Candidate Keys: {BookingID}

Checking each FD for BCNF:

- BookingID → UserID → BookingID is a Candidate Key
- BookingID → BookingDate → BookingID is a Candidate Key
- BookingID → BookingStatus → BookingID is a Candidate Key
- BookingID → PaidAmount → BookingID is a Candidate Key
- BookingID → PaymentStatus → BookingID is a Candidate Key
- BookingID → BusID → BookingID is a Candidate Key
- BookingID → DepartureDateAndTime → BookingID is a Candidate Key

Booking is in BCNF

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11) Trip

Attributes: BusID, DepartureDateAndTime, RouteID, JourneyStatus, DriverID, ConductorID

Finding Candidate Keys:

- {BusID, DepartureDateAndTime} → RouteID, JourneyStatus, DriverID, ConductorID → {BusID, DepartureDateAndTime} determines all attributes → **{BusID, DepartureDateAndTime}isaCandidateKey**

Candidate Keys: {BusID, DepartureDateAndTime}

Checking each FD for BCNF:

- {BusID, DepartureDateAndTime} → RouteID → {BusID, DepartureDateAndTime} is a Candidate Key
- {BusID, DepartureDateAndTime} → JourneyStatus → {BusID, DepartureDateAndTime} is a Candidate Key
- {BusID, DepartureDateAndTime} → DriverID → {BusID, DepartureDateAndTime} is a Candidate Key
- {BusID, DepartureDateAndTime} → ConductorID → {BusID, DepartureDateAndTime} is a Candidate Key

Trip is in BCNF